ThinkPower RS485 Protocol

Version:V01.05

1. **Summary**

The following defines the communication protocol between inverter and AP, which realizes the communication between inverter and AP through RS485, and is applicable to the 485 communication between ThinkPower on-grid inverter, hybrid Inverter and off grid Inverter

2. Communication interface

- AP is master，inverter is slave
- Referenced standards：MODBUS RTU
- Communicate Address：Device Addr:1-247;Brocast Addr:0;
  Inverter default addr:0X01
- Data structure
  Data：High byte + Low byte
  CRC：Low byte + High byte
- Interface parameters

Parameter	**Value**
Baud rate	9600 bps
Data length	8 bits
Check	No
Data flow control	No
Stop bit	1 bit

3. Application definition

MODBUS function codes defined in this agreement are as follows：

Code	MODBUS Definition	Data type	Inverter
0X03	Read register	word	Read inverter information, status, operation data and parameters
0X06	Write single register	word	Set parameters or control
0X10	Write multiple registers	word	Set parameters, time

Abnormal response code

Code	Definition
0X83	Abnormal response code

Abnormal response code definition：

Code	MODBUS Definition	Inverter
01	Illegal function	function code of the requested command is not allowed or CRC is checked incorrectly
02	Illegal data address	The request read/write address is out of range
03	Illegal data	The written value exceeds the upper or lower limits of data setting
04	Slave fail	06 command: slave command failed to be executed 16 command: the parameter setting fails

3.1 0x03Read Register

Host Read Command:

Device Addr	Function Code	Start Addr		Data length		CRC Checksum	
ID	【03H】	H Byte	L Byte	H Byte	L Byte	CRC L	CRC H
Slave Addr	0X03	Start Addr		Double byte (N)		Double byte	

Slave respond:

Device Addr	Function Code	Respond Data length	Data		CRC Checksum	
ID	【03H】	Length = 2*N	H Byte	L Byte	CRC L	CRC H
Slave Addr	0X03	Single byte	2* N bytes		Double byte	

Abnormal Respond:

Device Addr	Error Code	Respond Abnormal Type	CRC Checksum	
ID	【83H】	01/02/03/04	CRC L	CRC H
Slave Addr	0X83	0X01/0X02/0X03/0X04	Double byte	

3.2 0x06 Write Single Register Command

Host Write Command:

Device Addr	Function Code	Ouput Address		Ouput Data		CRC Checksum	
ID	【06H】	H Byte	L Byte	H Byte	L Byte	CRC L	CRC H
Slave Addr	0X06	Write Address		Write Data		Double byte	

Slave respond:

Device Addr	Function Code	Ouput Address		Data		CRC Checksum	
ID	【06H】	H Byte	L Byte	H Byte	L Byte	CRC L	CRC H
Slave Addr	0X06	Write Address		Write Data		Double byte	

Abnormal Respond:

Device Addr	Error Code	Respond Abnormal Type	CRC Checksum	
ID	【83H】	01/02/03/04	CRC L	CRC H
Slave Addr	0X83	0X01/0X02/0X03/0X04	Double byte	

3.3 0x10 Write Multiple Registers Command

Host Write Command:

Device Addr	Function Code	Start Addr		Data length		Number of bytes	Write Data		CRC Checksum	
ID	【10H】	H Byte	L Byte	H Byte	L Byte	2*N	H Byte	L Byte	CRC L	CRC H
Slave Addr	0X10	Start Addr		Double byte (N)		Single byte	N Double byte Data		Double byte	

Slave respond:

Device Addr	Function Code	Start Addr		Data length		CRC Checksum	
ID	【10H】	H Byte	L Byte	H Byte	L Byte	CRC L	CRC H
Slave Addr	0X10	Start Addr		Double byte (N)		Double byte	

Abnormal Respond:

Device Addr	Error Code	Respond Abnormal Type	CRC Checksum	
ID	【83H】	01/02/03/04	CRC L	CRC H
Slave Addr	0X83	0X01/0X02/0X03/0X04	Double byte	

4 Register definition

4.1 Version & Running Status Information

Version information							
Register Addr	Variable name	Definition	Length	Unit	Type	W/R	Note
30001	uwHardWare_Version	Inverter hardware ver	1	/	UINT16	R	
30002	uwFirmWare_Version	Inverter firmware ver	1	/	UINT16	R	
30003	uwBMS_HW_Version	BMS hardware ver	1	/	UINT16	R	
30004	uwBMS_FW_Version	BMS firmware ver	1	/	UINT16	R	
30005	uwBMS_Pro_Version	BMS protocol ver	1	/	UINT16	R	
30006	uw485Protocol_Version	Monitoring protocol ver	1	/	UINT16	R	
30007	uwInverterType	Inverter type	1	/	UINT16	R	0x00: Three phase hybrid inverter 0x01: Single phase hybrid inverter nverter 0x02: AC Coupler 0x03: Off grid inverter 0x04: Three phase on-grid inverter 0x05: Single phase on-grid inverter
30008	Inverter rated power	Inverter rated power	1	0.01kW	UINT16	R	
30009-30025	Reserve	保留	17				
Status information							
30026	uwRunMode	Running state	1	/	UINT16	R	0x00: Waitting 0x01:Work 0x02:Backup 0x03:Fault
30027	Reserve	Reserve	1				
30028	uwProtectCode1	Alarm code1	1	/	UINT16	R	Refer to Table 1
30029	uwProtectCode2	Alarm code 2	1	/	UINT16	R	
30030	uwProtectCode3	Alarm code 3	1	/	UINT16	R	
30031	uwProtectCode4	Alarm code 4	1	/	UINT16	R	
30032	uwProtectCode5	Alarm code 5	1	/	UINT16	R	
30033	uwProtectCode6	Alarm code 6	1	/	UINT16	R	

30034	Reserve	Reserve	1				
30035	uwBMSProtectCode1	BMS code1	1	/	UINT16	R	
30036	uwBMSProtectCode2	BMS code2	1	/	UINT16	R	
30037	uwBMSProtectCode3	BMS code3	1	/	UINT16	R	
30038	uwBMSProtectCode4	BMS code4	1	/	UINT16	R	
30039-30042	Reserve	Reserve	4				
30043	uwInvTemp	Temperature of inv part	1	0.1℃	UINT16	R	
30044	uwDcTemp	Temperature of dc part	1	0.1℃	UINT16	R	
30045-30050	Reserve	Reserve	6				

Tab1 Alarm code

Alarm code	Bit	Definition
uwProtectCode1	Bit15	Reserved
	Bit14	Reserved
	Bit13	Parallel comm line loss
	Bit12	MCU Internal comm failure
	Bit11	Bus soft start failure
	Bit10	Relay stick failure
	Bit9	Relay open failure
	Bit8	Grid Eps port reverse
	Bit7	DC Bus low
	Bit6	DC Bus high
	Bit5	DC Bus unbalance
	Bit4	Over temperature
	Bit3	Leakage current high
	Bit2	Isolation low
	Bit1	Bat volt high
	Bit0	PV volt high
uwProtectCode2	Bit15	Reserved
	Bit14	Reserved
	Bit13	Reserved
	Bit12	Reserved
	Bit11	Reserved
	Bit10	Reserved
	Bit9	Meter comm failure

	Bit8	Remote off
	Bit7	TOU mode no time setting
	Bit6	BMS comm failure
	Bit5	Bat SOC extremely low
	Bit4	Bat open
	Bit3	Bat volt low
	Bit2	Bat SOC low in EPS
	Bit1	Bat SOC low on Grid
	Bit0	Reserved
uwProtectCode3	Bit15	Reserved
	Bit14	Reserved
	Bit13	Reserved
	Bit12	Reserved
	Bit11	Reserved
	Bit10	Reserved
	Bit9	PLL failure
	Bit8	Reserved
	Bit7	Utility not three phase
	Bit6	DC inject high
	Bit5	Grid wave loss
	Bit4	Utility loss
	Bit3	Utility frequency low
	Bit2	Utility frequency high
	Bit1	Utility volt low
	Bit0	Utility volt high
uwProtectCode4	Bit15	Reserved
	Bit14	Reserved
	Bit13	Reserved
	Bit12	Reserved
	Bit11	Reserved
	Bit10	Reserved
	Bit9	Reserved
	Bit8	Reserved
	Bit7	Reserved
	Bit6	Reserved
	Bit5	Eps output short
	Bit4	Eps Inv soft start failure
	Bit3	Eps Bus soft start failure
	Bit2	Eps volt high
	Bit1	Eps volt low
	Bit0	Eps over load
uwProtectCode5	Bit15	Reserved
	Bit14	Reserved

	Bit13	Reserved
	Bit12	Reserved
	Bit11	Reserved
	Bit10	Reserved
	Bit9	Reserved
	Bit8	Reserved
	Bit7	Reserved
	Bit6	Reserved
	Bit5	Reserved
	Bit4	Reserved
	Bit3	Bus trip
	Bit2	Inv trip
	Bit1	Output open circuit
	Bit0	Bat trip
uwProtectCode6	Bit15	Reserved
	Bit14	Reserved
	Bit13	Reserved
	Bit12	Reserved
	Bit11	Reserved
	Bit10	Reserved
	Bit9	Reserved
	Bit8	Reserved
	Bit7	Reserved
	Bit6	Reserved
	Bit5	Reserved
	Bit4	Reserved
	Bit3	Reserved
	Bit2	Reserved
	Bit1	Reserved
	Bit0	Reserved

4.2 Inverter running data

PVInfo							
Register addr	Variable name	Definition	Length	Unit	Data Type	W/R	Note
31001	uwVpv1	PV1 volt	1	0.1V	UINT16	R	
31002	uwIpv1	PV1 curr	1	0.1A	UINT16	R	
31003	uwVpv2	PV2 volt	1	0.1V	UINT16	R	
31004	uwIpv2	PV2curr	1	0.1A	UINT16	R	

31005	uwVpv3	PV3 volt	1	0.1V	UINT16	R	
31006	uwIpv3	PV3 curr	1	0.1A	UINT16	R	
31007	uwPpv1	PV1power	1	W	UINT16	R	
31008	uwPpv2	PV2 power	1	W	UINT16	R	
31009	uwPpv3	PV3 power	1	W	UINT16	R	
31010	uwTotalPpv	Total PVpower	1	W	UINT16	R	
31011	uwVpbus	P Bus volt	1	0.1V	UINT16	R	
31012	uwVnbus	N Bus volt	1	0.1V	UINT16	R	
31013	uwVbus	Bus volt	1	0.1V	UINT16	R	
31014-31020	Reserve	Reserve	7				
Inv info							
31021	uwVac_R	Inv R volt	1	0.1V	UINT16	R	
31022	wIinv_R	Inv R curr	1	0.1A	INT16	R	
31023	uwFreq_R	Inv R freq	1	0.01Hz	UINT16	R	
31024	dwPinv_R_H	Inv R power H	1	W	INT16	R	
31025	dwPinv_R_L	Inv R power L	1	W	INT16	R	
31026	uwVac_S	Inv S volt	1	0.1V	UINT16	R	
31027	wIinv_S	Inv S curr	1	0.1A	INT16	R	
31028	uwFreq_S	Inv S freq	1	0.01Hz	UINT16	R	
31029	dwPinv_S_H	Inv S power H	1	W	INT16	R	
31030	dwPinv_S_L	Inv S power L	1	W	INT16	R	
31031	uwVac_T	Inv T volt	1	0.1V	UINT16	R	
31032	wIinv_T	Inv Tcurr	1	0.1A	INT16	R	
31033	uwFreq_T	Inv T freq	1	0.01Hz	UINT16	R	
31034	dwPinv_T_H	Inv T power H	1	W	INT16	R	
31035	dwPinv_T_L	Inv T power L	1	W	INT16	R	
31036	dwPinv_H	Inv Total power H	1	W	INT16	R	
31037	dwPinv_L	Inv Total power L	1	W	INT16	R	
31038	dwQinv_H	Inv Total reactive power H	1	W	INT16	R	
31039	dwQinv_L	Inv Total reactive power L	1	W	INT16	R	
31040	uwPfactor_inv	Inv power factor	1	0.01	UINT16	R	
31041-31050	Reserve	reserve	10				
Load info							
32001	uwVLoad_R	R phase load volt	1	0.1V	UINT16	R	
32002	uwVLoad_S	S phase load volt	1	0.1V	UINT16	R	
32003	uwVLoad_T	T phase load volt	1	0.1V	UINT16	R	
32004	dwPLoad_R_H	R phase load power H	1	W	INT16	R	

32005	dwPLoad_R_L	R phase load power L	1	W	INT16	R	
32006	dwPLoad_S_H	S phase load power H	1	W	INT16	R	
32007	dwPLoad_S_L	S phase load power L	1	W	INT16	R	
32008	dwPLoad_T_H	T phase load power H	1	W	INT16	R	
32009	dwPLoad_T_L	T phase load power L	1	W	INT16	R	
32010	dwPLoad_Total_H	Total load power H	1	W	INT16	R	
32011	dwPLoad_Total_L	Total load power L	1	W	INT16	R	
32012-32018	Reserve	Reserve	7				
Bat info							
32019	uwVbat	Battery volt	1	0.1V	UINT16	R	
32020	wIbat	Battery curr	1	0.1A	INT16	R	
32021	dwPbat_H	Battery power H	1	W	INT16	R	
32022	dwPbat_L	Battery power L	1	W	INT16	R	
32023	uwBatSOC	Battery SOC	1	0.1%	UINT16	R	
32024	uwBatSOH	Battery SOH	1	0.1%	UINT16	R	
32025	wBatTemp	Battery temperature	1	0.1℃	INT16	R	
32026	uwBatStatue	Battery statue	1		UINT16	R	0 Satandby 1 Discharge 2 Charge
32027	uwBatChgVoltEnd	Charge volt limit	1	0.1V	UINT16	R	
32028	uwBatDischgVoltEnd	Discharge volt limit	1	0.1V	UINT16	R	
32029	uwBatChgCurrMax	Maximum charge curr	1	0.1A	UINT16	R	
32030	uwBatDischgCurrMax	Maximum discharge curr	1	0.1A	UINT16	R	
32031	uwBmsStatue	Bms statue	1		UINT16	R	Different battery manufacturers have different definitions
32032	uwMaxCellVolt	Battery maximum cell volt	1	1mV	UINT16	R	
32033	uwMinCellVolt	Battery minimum cell volt	1	1mV	UINT16	R	
32034	wMaxCellTemp	Battery maximum cell temperature	1	0.1℃	INT16	R	
32035	wMinCellTemp	Battery minimum cell temperature	1	0.1℃	INT16	R	
32036	Reserve	Reserve	13				

32048							
Meter info							
33001	uwVac_Meter_R	R phase volt	1	0.1V	UINT16	R	
33002	wIac_Meter_R	R phase curr	1	0.1A	INT16	R	
33003	dwP_Meter_R_H	R phase power H	1	W	INT16	R	
33004	dwP_Meter_R_L	R phase power L	1	W	INT16	R	
33005	uwVac_Meter_S	S phase volt	1	0.1V	UINT16	R	
33006	wIac_Meter_S	S phase curr	1	0.1A	INT16	R	
33007	dwP_Meter_S_H	S phase power H	1	W	INT16	R	
33008	dwP_Meter_S_L	S phase power L	1	W	INT16	R	
33009	uwVac_Meter_T	T phase volt	1	0.1V	UINT16	R	
33010	wIac_Meter_T	T phase curr	1	0.1A	INT16	R	
33011	dwP_Meter_T_H	T phase power H	1	W	INT16	R	
33012	dwP_Meter_T_L	T phase power L	1	W	INT16	R	
33013	uwVac_Meter_RS	RS line volt	1	0.1V	UINT16	R	
33014	uwVac_Meter_ST	ST line volt	1	0.1V	UINT16	R	
33015	uwVac_Meter_TR	TR line volt	1	0.1V	UINT16	R	
33016	dwP_Meter_H	Grid total power H	1	W	INT16	R	
33017	dwP_Meter_L	Grid total power L	1	W	INT16	R	
33018-33031	Reserve	Reserve	14				
Power generation info							
34001	dwETotal_PV_H	Total PV generation H	1	0.1kWh	UINT16	R	
34002	dwETotal_PV_L	Total PV generation L	1	0.1kWh	UINT16	R	
34003	dwETotal_Chg_H	Battery total charge H	1	0.1kWh	UINT16	R	
34004	dwETotal_Chg_L	Battery total charge L	1	0.1kWh	UINT16	R	
34005	dwETotal_DisChg_H	Battery total discharge H	1	0.1kWh	UINT16	R	
34006	dwETotal_DisChg_L	Battery total discharge L	1	0.1kWh	UINT16	R	
34007	dwETotal_ToGrid_H	Grid total export H	1	0.1kWh	UINT16	R	
34008	dwETotal_ToGrid_L	Grid total export L	1	0.1kWh	UINT16	R	
3409	dwETotal_FromGrid_H	Grid total import H	1	0.1kWh	UINT16	R	
34010	dwETotal_FromGrid_L	Grid total import L	1	0.1kWh	UINT16	R	
34011	dwETotal_Load_H	Total load consumption H	1	0.1kWh	UINT16	R	
34012	dwETotal_Load_L	Total load consumption L	1	0.1kWh	UINT16	R	
34013	dwETotal_Inv_H	Inverter total	1	0.1kWh	UINT16	R	

		generation H					
34014	dwETotal_Inv_L	Inverter total generation L	1	0.1kWh	UINT16	R	
34015	dwEToday_PV_H	Today PV generation H	1	0.1kWh	UINT16	R	
34016	dwEToday_PV_L	Today PV generation L	1	0.1kWh	UINT16	R	
34017	dwEToday_Chg_H	Battery today charge H	1	0.1kWh	UINT16	R	
34018	dwEToday_Chg_L	Battery today charge L	1	0.1kWh	UINT16	R	
34019	dwEToday_DisChg_H	Battery today discharge H	1	0.1kWh	UINT16	R	
34020	dwEToday_DisChg_L	Battery today discharge L	1	0.1kWh	UINT16	R	
34021	dwEToday_ToGrid_H	Grid today export H	1	0.1kWh	UINT16	R	
34022	dwEToday_ToGrid_L	Grid today export L	1	0.1kWh	UINT16	R	
34023	dwEToday_FromGrid_H	Grid today import H	1	0.1kWh	UINT16	R	
34024	dwEToday_FromGrid_L	Grid today import L	1	0.1kWh	UINT16	R	
34025	dwEToday_Load_H	Today load consumption H	1	0.1kWh	UINT16	R	
34026	dwEToday_Load_L	Today load consumption L	1	0.1kWh	UINT16	R	
34027	dwEToday_Inv_H	Inverter today generation H	1	0.1kWh	UINT16	R	
34028	dwEToday_Inv_L	Inverter today generation L	1	0.1kWh	UINT16	R	
34029-34040	Reserve	Reserve	12				

4.3 Product&Parameter setting

ID info							
Register addr	**Variable name**	**Definition**	**Length**	**Unit**	**Data Type**	**W/R**	**Note**
40001-40008	sSerialNumber	Inverter ID	8	/	ASCII	R	ID fix16 number
40009-40016	sModelName	Inverter model	8	/	ASCII	R	The model length is uncertain, end with the '\0' return key
40017-40024	sBrandName	Inverter brand	8	/	ASCII	R	Brand uncertain, end with '\0' return key

Basic info							
40050	uwInverterRole	Inverter role (in parallel use)	1	/	UINT16	RW	0x00 Master 0x01 Slave
40051	uwModbus_ID	Inverter Modbus address	1	/	UINT16	RW	Default 0x01
40052	uwInternalCom_ID	Internal Modbus addr	1	/	UINT16	RW	Default 0x01
40053-40060	Reserve	Reserve	8				
Work mode setting							
41001	uwWorkMode	Inverter work mode	1	/	UINT16	RW	00 General mode 01 Peak shaving and valley filling mode 02 Battery backup mode 03 Micro grid mode
41002	uwAntiReverseEn	Anti-reverse enable	1	/	UINT16	RW	00 Disable 01 Enable
41003	uwReversePowerLimit	Export limit	1	%	UINT16	RW	0 forbid selling electricity 0-100
41004	uwACUnBalanceOutputEn	Three phase unbalance output	1	/	UINT16	RW	00 Disable 01 Enable
41005	uwOffTimeChgEn	Charge enable/disable in valley time	1	/	UINT16	RW	00 Disable 01 Enable
41006	uwOffTimeChgSOC	Charge aim soc in valley time	1	1%	UINT16	RW	20-100
41007	uwOffTimeChgPower	Charge power in valley time	1	W	UINT16	RW	>=500
41008	uwOffTime1En	Enable valley time period 1	1	/	UINT16	RW	00 Disable 01 Enable
41009	uwChgStartTime1_Hour	Valley time1start hour	1	/	UINT16	RW	0-24
41010	uwChgStartTime1_Minute	Valley time 1 start min	1	/	UINT16	RW	0-59
41011	uwChgEndTime1_Hour	Valley time1end hour	1	/	UINT16	RW	0-24
41012	uwChgEndTime1_Minute	Valley time 1 end min	1	/	UINT16	RW	0-59
41013	uwOffTime2En	Enable valley time period 2	1	/	UINT16	RW	00 Disable 01 Enable
41014	uwChgStartTime2_Hour	Valley time2start hour	1	/	UINT16	RW	0-24
41015	uwChgStartTime2_Minute	Valley time 2 start min	1	/	UINT16	RW	0-59
41016	uwChgEndTime2_Hour	Valley time2end hour	1	/	UINT16	RW	0-24
41017	uwChgEndTime2_Minute	Valley time 2 end min	1	/	UINT16	RW	0-59
41018	uwOffTime3En	Enable valley time period 3	1	/	UINT16	RW	00 Disable 01 Enable
41019	uwChgStartTime3_Hour	Valley time3start hour	1	/	UINT16	RW	0-24
41020	uwChgStartTime3_Minute	Valley time 3start min	1	/	UINT16	RW	0-59

41021	uwChgEndTime3_Hour	Valley time3end hour	1	/	UINT16	RW	0-24
41022	uwChgEndTime3_Minute	Valley time 3end min	1	/	UINT16	RW	0-59
41023	uwPeakTimeDischgMode	Discharge mode in peak time	1	/	UINT16	RW	00 follow smart meter 01 Fixed power
41024	uwPeakTimeDischgPower	Discharge power in peak time	1	W	UINT16	RW	>=500
41025	uwPeakTime1En	Enable peak time period 1	1	/	UINT16	RW	00 Disable 01 Enable
41026	uwDischgStartTime1_Hour	Peak time1start hour	1	/	UINT16	RW	0-24
41027	uwDischgStartTime1_Minute	Peak time 1 start min	1	/	UINT16	RW	0-59
41028	uwDischgEndTime1_Hour	Peak time1end hour	1	/	UINT16	RW	0-24
41029	uwDischgEndTime1_Minute	Peak time 1 end min	1	/	UINT16	RW	0-59
41030	uwPeakTime2En	Enable peak time period 2	1	/	UINT16	RW	00 Disable 01 Enable
41031	uwDischgStartTime2_Hour	Peak time2start hour	1	/	UINT16	RW	0-24
41032	uwDischgStartTime2_Minute	Peak time 2start min	1	/	UINT16	RW	0-59
41033	uwDischgEndTime2_Hour	Peak time2end hour	1	/	UINT16	RW	0-24
41034	uwDischgEndTime2_Minute	Peak time 2 end min	1	/	UINT16	RW	0-59
41035	uwPeakTime3En	Enable peak time period 3	1	/	UINT16	RW	00 Disable 01 Enable
41036	uwDischgStartTime3_Hour	Peak time3start hour	1	/	UINT16	RW	0-24
41037	uwDischgStartTime3_Minute	Peak time 3 start min	1	/	UINT16	RW	0-59
41038	uwDischgEndTime3_Hour	Peak time3end hour	1	/	UINT16	RW	0-24
41039	uwDischgEndTime3_Minute	Peak time 3 end min	1	/	UINT16	RW	0-59
41040	uwStorageChgEn	Charge enable/disable from grid in battery backup mode	1	/	UINT16	RW	00 Disable 01 Enable
41041	uwStorageChgSOC	Charge aim soc from grid in battery backup mode	1	1%	UINT16	RW	20-100
41042	uwStorageChgPower	Charge power in battery backup mode	1	W	UINT16	RW	>=500
41043-41050	Reserve	Reserve	8				
Battery Setting							
42001	uwBatType	Battery type	1	/	UINT16	RW	00 Lithium battery 01Lead acid battery

							02 No battery
42002	uwBMSProtocol	BMS protocol	1	/	UINT16	RW	Refer to table 2
42003	uwBatSocLowOnGrid	SOC discharge limit in on grid mode	1	1%	UINT16	RW	5-70
42004	uwBatSocLowOffLine	SOC discharge limit in off grid mode	1	1%	UINT16	RW	5-70
42005	uwLeadAcidBatCapAh	battery capacity	1	Ah	UINT16	RW	>=20
42006	uwLeadAcidBatMaxDischgCurr	Maximum battery discharge current	1	A	UINT16	RW	>=20
42007	uwLeadAcidBatMaxChgCurr	Maximum battery charging current	1	A	UINT16	RW	>=4
42008	uwLeadAcidBatChgFloatV	Battery floating charge voltage	1	0.1V	UINT16	RW	500-560
42009	uwLeadAcidBatDischgVoltShut	Battery discharge cut-off voltage	1	0.1V	UINT16	RW	400-480
42010	uwLeadAcidBatResistance	internal resistance of cell	1	mΩ	UINT16	RW	0-200
42011	uwLeadAcidBatVoltTempComp	Battery temperature compensation	1	mV/℃ /Cell	UINT16	RW	0-10
42012-42015	Reserve	Reserve	4				
Meter Protocol Setting							
42016	uwMeterProtocol	Meter protocol	1	/	UINT16	RW	00 No Meter 01 CHNT Three phase meter 02 CHNT Single phase meter 03 Acrel Three phase meter 04 Acrel Single phase meter 05 EastRon Three phase meter 06 EastRon Single phase meter
42017-42020	Reserve	Reserve	4				
EPS Setting							
42021	uwEPSEn	EPS Enable	1	/	UINT16	RW	00 Disable 01 Enable
42022	uwEPSVoltSet	EPS Volt	1	/	UINT16	RW	00: 208V 01: 220V 02: 230V

							03: 240V
42023	uwEPSFreqSet	EPS Freq	1	/	UINT16	RW	00:50Hz 01: 60Hz
42024- 42025	Reserve	Reserve	2				
Grid VHA Auto Low Power							
42026	uwGridVHAdjustEn	Grid volt high, inverter adjust power automatically	1	/	UINT16	RW	00 Disable 01 Enable
Buzzer setup							
42027	uwBuzEn	Buzzer Enable	1	/	UINT16	RW	00 Disable 01 Enable
RTC							
42028	dwRTC_H	RTC Counter H	1	/	UINT16		Based on 1970.1.1 00:00:00
42029	dwRTC_L	RTC Counter L	1	/	UINT16		Based on 1970.1.1 00:00:00
Shadow Function							
42030	uwShadowFuncEn	Shadow function enable	1	/	UINT16	RW	00 Disable 01 Enable
Isolation Function							
42031	uwISOFuncEn	Isolation Function enable	1	/	UINT16	RW	00 Disable 01 Enable
Energy monitoring (Only applicable to on-grid inverter)							
42032	uw24HourMonitorEn	Energy monitoring enable	1	/	UINT16	RW	00 Disable 01 Enable
AC Coupler							
42033	uwACCoupleFuncEn	AC Coupler enable	1	/	UINT16	RW	00 Disable 01 Enable
Parallel setup (only for single-phase energy storage inverters)							
42034	uwParallelMode	Parallel mode	1	/	UINT16	RW	00: Single-phase parallel 01: Three-phase parallel
42035	uwParallelPhase	The phase of the parallel	1	/	UINT16	RW	00: A phase 01: B phase 02:C phase
AFCI setting							
42036	AFCI function	AFCI	1	/	UINT16	RW	00 Disable 01 Enable
42037	Clear Arc fault	Clear AFCI alarm	1	/	UINT16	RW	Write 1 clear
42038- 42050	Reserve	Reserve	13				
Arrearage lock setting							
42051	uwOverDueLockEn	preinstall Arrearage	1	/	UINT16	RW	00 Disable

		lock					01 Enable
42052	uwCreditDays	preinstall Arrearage days	1	Day	UINT16	RW	10-365
42053	uwRemainDays	Remaining days of operation	1	Day	UINT16	RW	0-365
Smart load setting							
42101	Smart load function	Smart load function	1	/	UINT16	RW	00 Disable 01 Enable
42102	Smart load always on	Smart load manual operation	1	/	UINT16	RW	00: off 01: on
42103	Smart load start pv power	Smart load Starting PV power	1	W	UINT16	RW	0-65535
42104	Smart load start bat volt(Not suitable for lithium batteries)	Smart load Starting battery voltage	1	0.1V	UINT16	RW	0-10000
42105	Smart load end bat volt(Not suitable for lithium batteries)	Smart load Shutdown battery voltage	1	0.1V	UINT16	RW	0-10000
42106	Smart load start soc	Smart load start soc	1	1%	UINT16	RW	0-100
42107	Smart load end soc	Smart load end soc	1	1%	UINT16	RW	0-100
42108	Minimum time for a single run of smart load	Smart load Minimum time for a single run	1	min	UINT16	RW	0-1024
42109-42120	Reserve	Reserve	12				
Supplementary setting item							
42201	External CT power adjust	External CT power calibration	1	%	UINT16	RW	10-255
42202	Limit mode	Limit mode	1	/	UINT16	RW	00: Total power 01: Minimum phase power
42203	CT direction reverse	CT direction reverse	1	/	UINT16	RW	00 Disable 01 Enable
42204-42205	Maximum export power	Maximum export power	2	W	UINT32	RW	hihg16 at 42204， low 16 at 42205
42206-42220	Reserve	Reserve	15				
ATE testing (0x10 command not supported)							
42501	ATE Test	ATE Test	1	/	UINT16	RW	00：Normal 01：ATE Test 02：Aging mode
42502	PV1 Volt adjust	PV1 Volt adjust	1	/	UINT16	RW	2000-2096
42503	PV1 Current adjust	PV1 Current adjust	1	/	UINT16	RW	2000-2096
42504	PV2 Volt adjust	PV2 Volt adjust	1	/	UINT16	RW	2000-2096

42505	PV2Current adjust	PV2Current adjust	1	/	UINT16	RW	2000-2096
42506	PV3 Volt adjust	PV3 Volt adjust	1	/	UINT16	RW	2000-2096
42507	PV3 Current adjust	PV3 Current adjust	1	/	UINT16	RW	2000-2096
42508	PV4 Volt adjust	PV4 Volt adjust	1	/	UINT16	RW	2000-2096
42509	PV4 Current adjust	PV4 Current calibration	1	/	UINT16	RW	2000-2096
42510	Bat Volt adjust	Bat Volt adjust	1	/	UINT16	RW	2000-2096
42511	Bat Current adjust	Bat Current adjust	1	/	UINT16	RW	2000-2096
42512	Grid R phase Volt adjust	Grid R phase Volt adjust	1	/	UINT16	RW	2000-2096
42513	Grid R phase Current adjust	Grid R phase Current adjust	1	/	UINT16	RW	2000-2096
42514	Grid S phase Volt adjust	Grid S phase Volt adjust	1	/	UINT16	RW	2000-2096
42515	Grid S phase Current adjust	Grid S phase Current adjust	1	/	UINT16	RW	2000-2096
42516	Grid T phase Volt adjust	Grid T phase Volt adjust	1	/	UINT16	RW	2000-2096
42517	Grid T phase Current adjust	Grid T phase Current adjust	1	/	UINT16	RW	2000-2096
42518	Inv power adjust	Inv power adjust	1	/	UINT16	RW	2000-2096
42519	PV1 power adjust	PV1 power adjust	1	/	UINT16	RW	2000-2096
42520	PV2 power adjust	PV2 power adjust	1	/	UINT16	RW	2000-2096
42521	PV3 power adjust	PV3 power adjust	1	/	UINT16	RW	2000-2096
42522	PV4 power adjust	PV4 power adjust	1	/	UINT16	RW	2000-2096
42523	Bat power adjust	Bat power adjust	1	/	UINT16	RW	2000-2096
42524	Inv R DC bias Current adjust	ATE send current DC bias value to inverter	1	/	INT16	RW	[-1000,1000]
42525	Inv S DC bias Current adjust	ATE send current DC bias value to inverter	1	/	INT16	RW	[-1000,1000]
42526	Inv T DC bias Current adjust	ATE send current DC bias value to inverter	1	/	INT16	RW	[-1000,1000]
42527-42540	Reserve	Reserve	14				
Grid Setting							
43001	uwGridCode	Grid code	1	/	UINT16	RW	Refer to table 3
43002	uwReConnectTime	Reconnect time	1	S	UINT16	RW	Range 10-1000
43003	uwGridVoltHighLimit1	Grid Over volt limit1	1	0.1V	UINT16	RW	Range 2400-3000
43004	uwGridVoltLowLimit1	Grid Under volt limit1	1	0.1V	UINT16	RW	Range 1400-2200

43005	uwGridFreqHighLimit1	Grid Over freq limit1	1	0.01Hz	UINT16	RW	Range 5000-6500
43006	uwGridFreqLowLimit1	Grid Under freq limit1	1	0.01Hz	UINT16	RW	Range 4500-6000
43007	uwGridVolt10minLimit	10min Grid average over volt limit	1	0.1V	UINT16	RW	Range 2400-3000
43008	uwGridVolt10minEn	10min Grid average volt enable	1	/	UINT16	RW	0 Disable 1 Enable
43009	uwGridVoltHighLimit2	Grid Over volt limit 2	1	0.1V	UINT16	RW	Range 2400-3000
43010	uwGridVoltHighLimit3	Grid Over volt limit 3	1	0.1V	UINT16	RW	Range 2400-3000
43011	uwGridVoltLowLimit2	Grid Under volt limit 2	1	0.1V	UINT16	RW	Range 500-2200
43012	uwGridVoltLowLimit3	Grid Under volt limit 3	1	0.1V	UINT16	RW	Range 500-2200
43013	uwGridFreqHighLimit2	Grid Over freq limit 2	1	0.01Hz	UINT16	RW	Range 5000-6500
43014	uwGridFreqLowLimit2	Grid Under freq limit 2	1	0.01Hz	UINT16	RW	Range 4500-6000
43015	uwGridVoltHighTime1	Grid Over volt time1	1	0.01s	UINT16	RW	Range 0-65000
43016	uwGridVoltHighTime2	Grid Over volt time 2	1	0.01s	UINT16	RW	Range 0-65000
43017	uwGridVoltHighTime3	Grid Over volt time 3	1	0.01s	UINT16	RW	Range 0-65000
43018	uwGridVoltLowTime1	Grid Under volt time 1	1	0.01s	UINT16	RW	Range 0-65000
43019	uwGridVoltLowTime2	Grid Under volt time 2	1	0.01s	UINT16	RW	Range 0-65000
43020	uwGridVoltLowTime3	Grid Under volt time 3	1	0.01s	UINT16	RW	Range 0-65000
43021	uwGridFreqHighTime1	Grid Over Freq time1	1	0.01s	UINT16	RW	Range 0-65000
43022	uwGridFreqHighTime2	Grid Over Freq time 2	1	0.01s	UINT16	RW	Range 0-65000
43023	uwGridFreqLowTime1	Grid Under Freq time 1	1	0.01s	UINT16	RW	Range 0-65000
43024	uwGridFreqLowTime2	Grid Under Freq time 2	1	0.01s	UINT16	RW	Range 0-65000
43025	uwRocofFuncEn	Rocof enable	1	/	UINT16	RW	0 Disable 1 Enable
43026	uwRocofValue	Rocof Value	1	0.01Hz/s	UINT16	RW	Range 0-500
43027	uwLVRTFuncEn	LVRT enable	1	/	UINT16	RW	0 Disable 1 Enable
43028	uwLVRTVolt	LVRT volt	1	0.1V	UINT16	RW	Range 500-2200
43029	uwHVRTFuncEn	HVRT enable	1	/	UINT16	RW	0 Disable 1 Enable
43030	uwHVRTVolt	HVRT volt	1	0.1V	UINT16	RW	Range 2400-3000
43031	uwActiveIslandEn	Active Islanding enable	1	/	UINT16	RW	0 Disable 1 Enable
43032-43060	Reserve	Reserve	29				
Excute command							

44001	uwRemote_Off	Remote on/off	1	/	UINT16	RW	00: on 01: off
44002	uwResetRunData	Clear all running data	1	/	UINT16	W	Write 1 Execution
44003	uwResetPara	Restore all inverter parameters to factory defaults	1	/	UINT16	W	Write 1 Execution
44004	Battery Maintenance	Battery maintenance	1	/	UINT16	W	Write 1 Execution, together with 44005
44005	Battery maintenance soc	Battery maintenance capacity	1	%	UINT16	W	10-100
44006	Clear wait time	Clear latency	1	/	UINT16	W	Write 1 Execution
44007-44020	Reserve	Reserve	14				
并网安规类设置—有功功率控制							
45001	uwOverFreqDerateEn	Whether to enable over frequency power reduction	1	/	UINT16	RW	0 0:Disable 01 :Enable
45002	uwFreqOverLimitPStart	Start freq of over freq power reduction	1	0.01Hz	UINT16	RW	5020-6500
45003	uwFreqOverLimitPEnd	End freq of over freq power reduction	1	0.01Hz	UINT16	RW	5020-6500
45004	uwRate_PlimitbyFreq	Ratio of Over freq power reduction	%(Pref/Hz)	/	UINT16	RW	0-100
45005	uwStartPowerRate	Start power rate	1	%Pn/min	UINT16	RW	Range 0-100
45006	uwReconnectPowerRate	Reconnect power rate	1	%Pn/min	UINT16	RW	Range 0-100
45007	uwDRMFuncEn	DRM Function	1	/	UINT16	RW	0 0:Disable 01 :Enable
45008-45020	Reserve	Reserve	13				
并网安规类设置—无功功率控制							
46001	uwQManageEn	If enable Reactive power control	1	/	UINT16	RW	0 0:Disable 01 :Enable
46002	uwQCtrlMode	Reactive power control mode	1	/	UINT16	RW	00-----Fixed PF 01-----Fixed Q(Var) 02----CosPhi-P/Pn curve mode

							03-----Q(U) mode 04-----Pf mode
46003	uwFixedPF	Fixed Power Factor	1	0.001	INT16		Positive is lag reactive power; Negative is leading reactive power (-1000, -800]∪[800, 1000]
46004	dwFixQpower_H	Fixed Reactive Power Data H	1	VA	INT16		Positive is lag reactive power; Negative is leading reactive power
46005	dwFixQpower_L	Fixed Reactive Power Data L	1	VA	INT16		
46006	uwCosPhiP_P1	1st point P/Pn of CosPhi-P/Pn curve	1	%Pn	UINT16		[0-100]
46007	uwCosPhiP_CosPhi1	1st point CosPhi of CosPhi-P/Pn curve	1	0.001	INT16		Positive is lag reactive power; Negative is leading reactive power (-1000, -800]∪[800, 1000]
46008	uwCosPhiP_P2	2nd point P/Pn of CosPhi-P/Pn curve	1	%Pn	UINT16		[0-100]
46009	uwCosPhiP_CosPhi2	2nd point CosPhi of CosPhi-P/Pn curve	1	0.001	INT16		Positive is lag reactive power; Negative is leading reactive power (-1000, -800]∪[800, 1000]
46010	uwQUStart_P	Q(U) mode trigger power P/Pn	1	%Pn	UINT16		[0-100]
46011	uwQUEnd_P	Q(U) mode cut-off power P/Pn	1	%Pn	UINT16		[0-100]
46012	uwQU_U1	control 1st point U/Un of Q(U) curve	1	Un%	UINT16		[80-100]
46013	uwQU_Q1	control 1st point Q/Pn of Q(U) curve	1	0.1Pn%	INT16		[0, 483]
46014	uwQU_U2	control 2nd point U/Un of Q(U) curve	1	Un%	UINT16		[80-100]
46015	uwQU_Q2	control 2nd point Q/Pn of Q(U) curve	1	0.1Pn%	INT16		[0, 483]
46016	uwQU_U3	control 3rd point U/Un of Q(U) curve	1	Un%	UINT16		[100-136]
46017	uwQU_Q3	control 3rd point Q/Pn	1	0.1Pn%	INT16		[-483, 0]

		of Q(U) curve					
46018	uwQU_U4	control 4th point U/Un of Q(U) curve	1	Un%	UINT16		[100-136]
46019	uwQU_Q4	control 4th point Q/Pn of Q(U) curve	1	0.1Pn%	INT16		[-483，0]
46020-46050	Reserve	Reserve	31				

Table2 BMSprotocol

uwBMSProtocol	BMS protocol
0x00	No BMS Protocol
0x01	TSolar High Voltage battery
0x02	TSolar Low Voltage battery
0x03	PYLON High Voltage battery
0x04	PYLON Low Voltage battery
0x05	GW High Voltage battery
0x06	GW Low Voltage battery
0x07	SOLTARO High Voltage battery
0x08	SOLTARO Low Voltage battery
0x09	Dyness High Voltage battery
0x0A	Dyness Low Voltage battery
0x0B	UHome Battery
0x0C	Camel Battery
0x0D	BYD Battery
0x0E	WOTAI High Voltage Battery
0x0F	WOTAI Low Voltage Battery
0x10	RUIPU Battery
0x11	Eagle High Voltage Battery
0x12	Eagle Low Voltage Battery
0x13	Hongzheng High Voltage Battery
0x14	Hongzheng Low Voltage Battery

Table3 Grid code

uwBMSProtocol	Grid code
0x00	50Hz Standard Grid
0x01	60Hz Standard Grid
0x02	Germany
0x03	Australia
0x04	United Kingdom

0x05	China
0x06	Italy
0x07	France
0x08	South Africa
0x09	Czech Republic
0x0A	Austria
0x0B	Belgium
0x0C	Switzerland
0x0D	Philippines
0x0E	Denmark
0x0F	Thailand
0x10	Greece
0x11	Netherland
0x12	India
0x13	New Zealand
0x14	Ireland
0x15	Poland
0x16	Spain
0x17	Romania